Patterns, proofs & practice

A mixed problem set for algebra, geometry, and early calculus

Name _____ Date _____

1. Rational equations

Solve the equation. State the excluded value and check your solution.

$$\frac{3x - 2}{x + 1} = 2 + \frac{1}{x + 1} \qquad x \neq -1$$

2. Quadratic structure

Factor the function, identify its zeros, and write it in completed-square form.

$$f(x) = x^2 - 6x + 5$$

Zeros: _____ Vertex: _____

3. A line through two points

The diagram shows A(−2, 1) and B(3, 4).

(a) Find the slope of line AB.
(b) Write an equation of line AB.
(c) Find its y-intercept.

